

Project Assignment 2024-25

Projects can be done by groups with no more than 3 members (3 is recommended). Group of a single student is allowed only on motivated cases.

Each group should choose a project topic (different groups can choose the same).

Each group will implement the project and produce a small report and present the work with a PowerPoint presentation (**15 minutes**).

During the presentation every member will discuss a part of the project.

Every member should be able to implement small modifications on the fly during the presentation.

There is only 1 day per exam round, on which there will be both project presentations and orals. Students can choose to present the project during a session and perform the oral part on a later session. To perform the oral part, it is necessary to have a positive evaluation of the project.

Operative Steps

1. Choose the project topic and send an email to maurizio.palmieri@unipi.it with subject [DSSProject2025]. The body of the email should contain the chosen topic and the names of the group members. Include all the members in cc. Material needed for the project will be provided in the reply.
2. (Optional) Schedule via email a meeting to discuss about initial design choices.
3. (Optional) After approximately 2/3 of the project, schedule a meeting to review the current state and initial results.
4. All the members of the project must enrol to the exam. (Optional) Students can specify if they are willing to present the project and perform the oral on another session.
5. Submission of the project:
 - The project must be submitted via email to **both** teachers at least two days before exam.
 - The code, the documentation and the presentation must be attached to the email.
6. Present the project within the time limit and answer the related questions.

Project Topic 1

Analyze the malicious payload in the different samples of APKs.

The report must include the following sections:

- Anti malware analysis: by submitting the samples to the tool web VirusTotal and discussing the result.
- Static analysis: by using Bytecodeviewer or JD-GUI to highlight interesting snippet of code (include and discuss such code in the report) and by using MobSF framework (<https://github.com/MobSF/Mobile-Security-Framework-MobSF>) to extract automatically features from the code interesting for malware analysis.
- (Optional) Dynamic analysis: execute the application in a controlled environment (a simulator) and observe the behavior of the application.
- The APKs will be provided on request (There is a small amount of available APK...)

This project does not involve the submission of the code.

Project Topic 2

Analyze a co-simulation scenario with a platoon of vehicles following a leader car.

- Analyze the scenario without attacks, running 3 co-simulations with different parameter values with the INTO-CPS Application,
- Inject an attack that forces a byzantine behaviour of the leader (sending 2 different values for speed).
- Implement the Oral Message consensus algorithm (OM1) among the vehicles and run again 1 co-simulation with 1 possible malicious behaviour of the leader and analyze the result.

The related FMUs, with their instructions, will be provided on request

Project Topic 3

Create an ADVISE model on Mobius for adversaries trying to attack the external connectivity and connection of a car, using the threats suggested in the table next page.

- Create an attack graph with roughly 20 elements (knowledge, access, skill, attack step and goal).
- Create at least 2 different adversary profiles, varying initial elements and skills.
- Evaluate the probability of each adversary to achieve the goal of blocking the car.
- Analyze the results.
- In the report, explicitly link the elements of the AEG with the elements of the table.

Table taken from **UN Regulation No. 155 - Cyber security and cyber security management system**

<https://unece.org/transport/documents/2021/03/standards/un-regulation-no-155-cyber-security-and-cyber-security>

<i>High level and sub-level descriptions of vulnerability/ threat</i>			<i>Example of vulnerability or attack method</i>	
4.3.5 Threats to vehicles regarding their external connectivity and connections	16	Manipulation of the connectivity of vehicle functions enables a cyber-attack, this can include telematics; systems that permit remote operations; and systems using short range wireless communications	16.1	Manipulation of functions designed to remotely operate systems , such as remote key, immobilizer, and charging pile
			16.2	Manipulation of vehicle telematics (e.g. manipulate temperature measurement of sensitive goods, remotely unlock cargo doors)
			16.3	Interference with short range wireless systems or sensors
	17	Hosted 3rd party software, e.g. entertainment applications, used as a means to attack vehicle systems	17.1	Corrupted applications , or those with poor software security, used as a method to attack vehicle systems
	18	Devices connected to external interfaces e.g. USB ports, OBD port, used as a means to attack vehicle systems	18.1	External interfaces such as USB or other ports used as a point of attack, for example through code injection
			18.2	Media infected with a virus connected to a vehicle system
			18.3	Diagnostic access (e.g. dongles in OBD port) used to facilitate an attack, e.g. manipulate vehicle parameters (directly or indirectly)

Project Topic 4

Model, as a SAN, a set of indoor temperature sensors under the following assumptions:

1. Each sensor outputs a boolean value $[0, 1]$:
 - a. '1' if the temperature is above the operative threshold,
 - b. '0' if the temperature is below the operative threshold.
2. The sensors have 3 possible states:
 - a. Properly functioning,
 - b. Wrong output,
 - c. Broken.
3. At the beginning, all the sensors are properly functioning.

Project Topic 4 (cont)

1. A “properly working” sensor has a fixed probability ($p_f = 0.05$) of switching to “wrong output” state every hour,
2. A “wrong output” sensor has a fixed probability ($p_r = 0.1$) of switching to “properly working” state every half an hour,
3. Each sensor always has a possibility, with exponential distribution with fixed rate ($\lambda = 0.01$ per hour) to end up in the broken state,
 - a. A broken sensor always outputs ‘0’,
 - b. There is no detection system (i.e. it is not possible to discern if the output ‘0’ is related to low temperature or a failure),
 - c. There is no recovery system (once broken the sensor stays broken forever).

Project Topic 4 (cont)

Considering that the system composed of many sensors is available as long as the majority of the sensors is providing the correct output.

Evaluate, using the simulation, the availability of the system every half an hour for 1 week under the condition that

1. the temperature is always higher than the threshold,
2. the temperature is always lower than the threshold.

Study the behaviour of the system with different values of p_f , p_r , λ and different number of sensors.